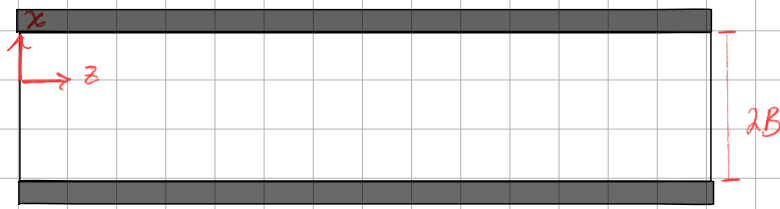


Problem set 4:-



Assumptions: constant ρ and μ .

at $t=0$ pressure gradient $\frac{\Delta P}{L} = \frac{P(z=L) - P(z=0)}{L}$ sets the fluid in motion.

① what direction and what components?

$$v_x = v_y = 0 \quad v_z \neq 0$$

$$v_z(y) = v_z(z) = 0 \quad v_z(x) \neq 0 \quad v_z(t) \neq 0.$$

using Equations of motion for a Newtonian fluid at constant ρ and μ : Appendix B:-
using (B.6-3) in cartesian coordinates

$$\rho \left(\frac{\partial v_z}{\partial t} + v_x \frac{\partial v_z}{\partial x} + v_y \frac{\partial v_z}{\partial y} + v_z \frac{\partial v_z}{\partial z} \right) = -\frac{\partial P}{\partial z} + \mu \left(\frac{\partial^2 v_z}{\partial x^2} + \frac{\partial^2 v_z}{\partial y^2} + \frac{\partial^2 v_z}{\partial z^2} \right) + \rho g_z$$

Annotations: $v_x=0$, $v_y=0$, $v_z \neq v_z(z)$, $v_z \neq v_z(y)$, $v_z \neq v_z(z)$.

$$\rho \frac{\partial v_z}{\partial t} = -\frac{\partial P}{\partial z} + \mu \frac{\partial^2 v_z}{\partial x^2} \rightarrow \rho \frac{\partial v_z}{\partial t} = \mu \frac{\partial^2 v_z}{\partial x^2} - \frac{\Delta P}{L}$$

② $\frac{\partial v_z}{\partial t} = 0$ $\mu \frac{\partial^2 v_z}{\partial x^2} = \frac{\Delta P}{L}$

integrate

$$\frac{dv_z}{dx} = \frac{\Delta P}{\mu L} x + C_1$$

integrate

$$v_z(x) = \frac{\Delta P}{2\mu L} x^2 + C_1 x + C_2$$

Applying Boundary conditions:-

$$v_z(x=-B) = 0$$

$$v_z(x=B) = 0$$

$$\text{at } x=0 \quad v_z = v_{\max}$$

$$v_z(x=B)=0$$

$$0 = \frac{\Delta P}{2\mu L} B^2 + C_1(B) + C_2 \quad \text{--- (1)}$$

$$v_z(x=-B)=0$$

$$0 = \frac{\Delta P}{2\mu L} (-B)^2 + C_1(-B) + C_2 \quad \text{--- (2)}$$

Adding both equations.

$$0 = \frac{2\Delta P}{2\mu L} B^2 + 0 + 2C_2$$

$$C_2 = -\frac{\Delta P}{2\mu L} B^2$$

$$0 = \frac{\Delta P}{2\mu L} B^2 + BC_1 - \frac{\Delta P B^2}{2\mu L}$$

$$BC_1 = 0$$

$$C_1 = 0$$

$$v_z(x) = \frac{\Delta P}{2\mu L} x^2 + \cancel{C_1}x + C_2 \quad \text{--- (0)}$$

$$v_z(x) = \frac{\Delta P}{2\mu L} x^2 - \frac{\Delta P}{2\mu L} B^2$$

$$= \frac{\Delta P}{2\mu L} (x^2 - B^2)$$

$$= -\frac{B^2 \Delta P}{2\mu L} \left(1 - \frac{x^2}{B^2}\right)$$

$$= -\frac{B^2 \Delta P}{2\mu L} \left(1 - \left(\frac{x}{B}\right)^2\right)$$

$$\text{at } x=0 \therefore v_z(x=0) = -\frac{B^2 \Delta P}{2\mu L} (1-0)$$

$$v_z(x=0) = v_{\max} = -\frac{B^2 \Delta P}{2\mu L}$$



③ $\frac{\partial \phi}{\partial \tau} = \frac{\partial^2 \phi}{\partial \eta^2} + 2.$

↓ dimensionless velocity $\frac{u_z}{u_{max}}$

The partial diff. eq: $\rho \frac{\partial u_z}{\partial t} = \mu \frac{\partial^2 u_z}{\partial x^2} - \frac{\Delta p}{L}$

$\eta = \eta(x, t)$

$u_z = u_z(x, t) = u_z(\eta(x, t)) = u_z(\eta).$

$\rho \frac{du_z}{dt} = \mu \frac{d^2 u_z}{dx^2} - \frac{\Delta p}{L}$

$\rho \frac{\partial u_z}{\partial t} = \mu \frac{\partial^2 u_z}{\partial x^2} - \frac{\Delta p}{L}$

$\frac{\partial u_z}{\partial t} = \left(\frac{\mu}{\rho} \right) \frac{\partial^2 u_z}{\partial x^2} - \frac{\Delta p}{\rho L}$

kinematic viscosity ν

$\phi = \frac{u_z}{u_{max}} \rightarrow u_z(x, t) = u_{max} \phi(\eta, \tau).$

$\frac{\partial u_z}{\partial t} = u_{max} \frac{\partial \phi}{\partial t} = u_{max} \frac{\partial \phi}{\partial \tau} \times \frac{d\tau}{dt}$

we know: $\tau = \frac{\mu t}{\rho B^2}$

$\frac{d\tau}{dt} = \frac{\mu}{\rho B^2}$

$\frac{\partial u_z}{\partial t} = \frac{u_{max} \mu}{\rho B^2} \frac{\partial \phi}{\partial \tau}$

$\frac{\partial^2 u_z}{\partial x^2} = u_{max} \frac{\partial^2 \phi}{\partial \eta^2} \times \left(\frac{d\eta}{dx} \right)^2$

$\eta = \frac{x}{B}$

$\frac{d\eta}{dx} = \frac{1}{B} \rightarrow \frac{d^2 \eta}{dx^2} = \frac{1}{B^2}$

$\frac{\partial u_z}{\partial x} = \frac{u_{max}}{B} \frac{\partial \phi}{\partial \eta}$

From u_{max} : $u_{max} = -B^2 \left(\frac{\Delta p}{2 \mu L} \right)$

$\frac{\Delta p}{L} = \frac{2 u_{max} \mu}{-B^2}$

Now combine all the terms together:-

$\frac{u_{max} \mu}{\rho B^2} \frac{\partial \phi}{\partial \tau} = \frac{\mu}{\rho} \frac{u_{max}}{B^2} \frac{\partial^2 \phi}{\partial \eta^2} + \frac{2 u_{max} \mu}{\rho B^2}$

$$\frac{d\phi}{d\tau} = \frac{d^2\phi}{d\eta^2} + 2$$

④

$$\frac{d\phi}{d\tau} = \frac{d^2\phi}{d\eta^2} + 2.$$

$$\text{@ S.S } \frac{d\phi}{d\tau} = 0.$$

$$\int \frac{d^2\phi}{d\eta^2} = \int -2$$

$$\int \frac{d\phi}{d\eta} = \int -2\eta + C_1$$

$$\phi(\eta) = -\eta^2 + C_1\eta + C_2.$$

Boundary Conditions:-

$$x=B \rightarrow Vz=0$$

$$x=-B \rightarrow Vz=0$$

$$\eta = \frac{x}{B}$$

$$x=B \rightarrow \eta = 1$$

$$x=-B \rightarrow \eta = -1$$

$$\phi(\eta) = \frac{Vz}{v_{max}}$$

$$\eta = 1 \rightarrow \phi(\eta) = 0$$

$$\eta = -1 \rightarrow \phi(\eta) = 0$$

$$\text{@ } \eta = 1 \quad \phi(1) = -1 + C_1 + C_2 = 0$$

$$\text{@ } \eta = -1 \quad \phi(-1) = -1 - C_1 + C_2 = 0$$

$$-2 + 2C_2 = 0$$

$$2C_2 = 2$$

$$C_2 = 1$$

$$C_1 + C_2 = 1$$

$$C_1 + 1 = 1$$

$$C_1 = 0$$

$$\phi(\eta) = -\eta^2 + 0 + 1$$

$$\phi(\eta) = 1 - \eta^2$$

$$\textcircled{5} \quad \phi(\eta, \tau=0) = 0$$

$$\phi(\eta=-1, \tau) = 0$$

$$\phi(\eta=1, \tau) = 0$$

$$\textcircled{6} \quad \phi(\eta, \tau) = \sum_{n=1}^{\infty} C_n(\tau) \cos\left(\frac{2n-1}{2} \pi \eta\right)$$

$$\frac{\partial \phi}{\partial \tau} = \sum_{n=1}^{\infty} \frac{\partial C_n}{\partial \tau} \cos(K_n \eta)$$

$$\frac{\partial \phi}{\partial \eta} = \sum_{n=1}^{\infty} C_n \times -K_n \sin(K_n \eta)$$

$$\frac{\partial^2 \phi}{\partial \eta^2} = \sum_{n=1}^{\infty} C_n \times -K_n^2 \cos(K_n \eta)$$

$$\frac{\partial \phi}{\partial \tau} = \frac{\partial^2 \phi}{\partial \eta^2} + 2$$

$$\sum_{n=1}^{\infty} \left[\frac{\partial C_n}{\partial \tau} \cos(K_n \eta) \right] = \sum_{n=1}^{\infty} \left[(-K_n)^2 C_n \cos(K_n \eta) \right] + 2$$

$$\sum_{n=1}^{\infty} \left[\frac{\partial C_n}{\partial \tau} \cos(K_n \eta) + K_n^2 C_n \cos(K_n \eta) \right] - 2 = 0 \quad || \cos(K_n \eta)$$

$$\sum_{n=1}^{\infty} \left[\cos(K_n \eta) \cdot \cos(K_n \eta) \left(\frac{\partial C_n}{\partial \tau} + K_n^2 C_n \right) \right] - 2 \cos(K_n \eta) = 0$$

now integrate $-1 \rightarrow 1$

$$\sum_{n=1}^{\infty} \int_{-1}^1 \cos(K_n \eta) \cdot \cos(K_n \eta) \left(\frac{\partial C_n}{\partial \tau} + K_n^2 C_n \right) d\eta - 2 \int_{-1}^1 \cos(K_n \eta) d\eta = 0$$

$$\frac{a}{\pi} \int_{-\frac{\pi}{a}}^{\frac{\pi}{a}} \cos(nax) \cos(nax) dx = \delta_{nm} = 1 \quad a = \pi$$

$$\frac{\partial C_n}{\partial \tau} + K_n^2 C_n - 2 \int_{-1}^1 \cos(K_n \eta) d\eta = 0$$

$$\frac{-2}{K_n} \sin(K_n \eta) \Big|_{-1}^1 = \frac{-2}{K_n} (\sin(K_n) + \sin(K_n))$$

$$-\frac{4}{K_n} (-1)^{n-1} = -4 \left(\frac{2}{(2n-1)\pi} \right) (-1)^{n-1} = -\frac{8}{\pi (2n-1)} (-1)^{n-1}$$

$$\frac{dc_n}{d\tau} + kn^2 c_n - \frac{4}{kn} (-1)^{n-1} = 0$$

$$c_n(\tau) = \frac{4}{kn^3} (-1)^{n-1} + A e^{-kn^2 \tau}$$

Boundary conditions:-

$$\phi(\eta, \tau=0) = 0$$

$$\phi(\eta, 0) = \sum_{n=1}^{\infty} c_n(0) \cos(k_n \eta) = 0$$

$$c_n(0) = \frac{4}{kn^3} (-1)^{n-1} + A e^{0}$$

$$0 = \sum_{n=1}^{\infty} \left(\frac{4}{kn^3} (-1)^{n-1} + A \right) \cos(k_n \eta)$$

$$A = -\frac{4}{kn^3} (-1)^{n-1}$$

$$c_n(\tau) = \frac{4}{kn^3} (-1)^{n-1} + \frac{4}{kn^3} (-1)^{n-1} e^{-kn^2 \tau}$$

$$c_n(\tau) = \frac{4}{\left(\frac{2n-1}{2}\pi\right)^3} (-1)^{n-1} + \frac{4}{\left(\frac{2n-1}{2}\pi\right)^3} (-1)^{n-1} e^{-\left(\frac{2n-1}{2}\pi\right)^2 \tau}$$

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